

## 40.3 Integrated likelihood approximations

Everything the last section promised rests on the integral Eq. 40.2.6, which has no closed form except in the normal linear case. Fitting a GLMM is therefore a numerical problem before it is a statistical one, and the choice of method is not innocent: three methods in common use can differ in the second significant figure of  $\hat{\beta}$  and in the first of  $\hat{\tau}$ .

### 40.3.1 Why it is hard, and what makes it easier

For the normal linear mixed model  $\mathbf{Y} \mid \mathbf{u}$  and  $\mathbf{u}$  are both normal, so Eq. 40.2.6 is a Gaussian integral and gives Eq. 32.2.2 in closed form. Nothing of the sort happens otherwise: a Poisson likelihood in  $e^u$  against a normal density in  $u$  is not a recognizable integral in any dimension. What *does* help is independence. If the random effects are attached to clusters that are independent of each other, the  $q$ -dimensional integral splits into  $m$  integrals of dimension  $q_0$  each, and with a random intercept  $q_0 = 1$ . A one-dimensional integral repeated  $m$  times is tractable; a  $q$ -dimensional one with  $q$  in the hundreds is not.

#### Theorem 40.3.1. The integrated likelihood and its approximations

In Definition 40.2.1 write  $\psi_i(\mathbf{u}_i) = \log f(\mathbf{y}_i \mid \mathbf{u}_i; \boldsymbol{\beta}, \phi) + \log p(\mathbf{u}_i; \boldsymbol{\theta})$ .

- (a) **Factorization.** If the random effects split into independent blocks  $\mathbf{u}_1, \dots, \mathbf{u}_m$  with  $\mathbf{u}_i$  entering only the observations of cluster  $i$ , then

$$L(\boldsymbol{\beta}, \boldsymbol{\theta}, \phi) = \prod_{i=1}^m \int_{\mathbb{R}^{q_0}} \exp\{\psi_i(\mathbf{u}_i)\} d\mathbf{u}_i, \quad (40.3.1)$$

a product of  $m$  integrals of dimension  $q_0$ .

- (b) **Laplace approximation.** Fix  $i$  and take  $q_0 = 1$ . Let  $\psi_i$  be four times continuously differentiable with a unique interior maximizer  $\hat{u}_i$ , and  $c_i = -\psi_i''(\hat{u}_i) > 0$ . Then, formally,

$$\begin{aligned} \int e^{\psi_i(u)} du &= \sqrt{\frac{2\pi}{c_i}} e^{\psi_i(\hat{u}_i)} \{1 + \delta_i + \dots\}, \\ \delta_i &= \frac{\psi_i^{(4)}(\hat{u}_i)}{8c_i^2} + \frac{5\{\psi_i^{(3)}(\hat{u}_i)\}^2}{24c_i^3}. \end{aligned} \quad (40.3.2)$$

Suppose  $\psi_i = n_i k_i + s_i$ , where  $k_i$  does not depend on  $n_i$  and has a nondegenerate maximum and  $s_i$  has derivatives bounded uniformly in  $n_i$ . (In a GLMM  $n_i k_i$  is the conditional log-likelihood and  $s_i$  the log prior density of  $\mathbf{u}_i$ , which does not grow with  $n_i$ .) Then both terms of  $\delta_i$  are  $O(n_i^{-1})$ : the Laplace approximation has *relative* error  $O(n_i^{-1})$  and the error in the log-likelihood is  $O(\sum_i n_i^{-1})$ .

- (c) **Adaptive Gauss–Hermite quadrature.** Let  $(z_k, w_k)_{k=1}^K$  be the nodes and weights of  $K$ -point Gauss–Hermite quadrature, so that  $\int e^{-z^2} P(z) dz = \sum_k w_k P(z_k)$  for every polynomial  $P$  of degree at most  $2K - 1$ . With  $\hat{\sigma}_i = c_i^{-1/2}$ , the substitution  $u = \hat{u}_i + \sqrt{2} \hat{\sigma}_i z$  gives the approximation

$$\int e^{\psi_i(u)} du \approx \sqrt{2} \hat{\sigma}_i \sum_{k=1}^K w_k e^{z_k^2} e^{\psi_i(\hat{u}_i + \sqrt{2} \hat{\sigma}_i z_k)}. \quad (40.3.3)$$

It is exact whenever  $e^{\psi_i(\hat{u}_i + \sqrt{2} \hat{\sigma}_i z) + z^2}$  is a polynomial of degree at most  $2K - 1$  in  $z$ ,

and  $K = 1$  reproduces [Eq. 40.3.2](#) without its correction terms.

- (d) **Penalized quasi-likelihood** avoids the integral altogether, by a penalized maximization that is cheaper than either of the above and carries a bias that does not vanish. It is stated and proved separately as [Proposition 40.3.2](#).

*Proof.*

- (a) Under the stated independence the integrand factorizes and Fubini's theorem applies, every factor being nonnegative.
- (b) Substitute  $u = \hat{u}_i + s/\sqrt{c_i}$  and expand  $\psi_i$  about  $\hat{u}_i$ , using  $\psi'_i(\hat{u}_i) = 0$ :

$$\begin{aligned}\psi_i(u) &= \psi_i(\hat{u}_i) - \frac{1}{2}s^2 + a_3s^3 + a_4s^4 + \dots, \\ a_3 &= \frac{\psi_i'''(\hat{u}_i)}{6c_i^{3/2}}, \quad a_4 = \frac{\psi_i^{(4)}(\hat{u}_i)}{24c_i^2}.\end{aligned}$$

Hence  $\int e^{\psi_i} du = c_i^{-1/2} e^{\psi_i(\hat{u}_i)} \int e^{-s^2/2 + a_3s^3 + a_4s^4 + \dots} ds$ . Expanding the second exponential as  $1 + (a_3s^3 + a_4s^4) + \frac{1}{2}a_3^2s^6 + \dots$  and using the standard normal moments  $ES^3 = 0$ ,  $ES^4 = 3$ ,  $ES^6 = 15$  gives the factor  $\sqrt{2\pi} \{1 + 3a_4 + \frac{15}{2}a_3^2 + \dots\}$ , which is [Eq. 40.3.2](#). If  $\psi_i = n_i k_i + s_i$  then  $c_i = n_i |k_i''| + O(1)$ ,  $\psi_i''' = n_i k_i''' + O(1)$  and  $\psi_i^{(4)} = n_i k_i^{(4)} + O(1)$ , so  $a_4 = O(n_i^{-1})$  and  $a_3^2 = O(n_i^{-1})$ .

- (c) The substitution is a change of variable, and  $e^{\psi_i(u)} = e^{-z^2} \{e^{z^2} e^{\psi_i(\hat{u}_i + \sqrt{2}\hat{\sigma}_i z)}\}$ , so the quadrature rule applies to the term in braces. For  $K = 1$  the node is  $z_1 = 0$  and the weight is  $w_1 = \sqrt{\pi}$ , giving  $\sqrt{2}\hat{\sigma}_i \sqrt{\pi} e^{\psi_i(\hat{u}_i)} = \sqrt{2\pi/c_i} e^{\psi_i(\hat{u}_i)}$ . ■

*Remark* (What “formally” means in (b)). The expansion in the proof interchanges an infinite series with an integral over the whole line, which the argument does not justify, and the remainder can also be affected by the behaviour of  $\psi_i$  far from its mode. The rigorous statement — that the expansion is asymptotic in  $n_i$ , with an explicit remainder, under conditions on the derivatives of  $k_i$  and on the tails — is classical, and is given in the form used here by Tierney and Kadane (1986) and Barndorff-Nielsen and Cox (1989, chapter 3). [Chapter 39](#) proves the companion statement for the posterior and the marginal likelihood (Theorem 39.4.2). We take the error order as given; [Example \(Three methods on clustered binary data\)](#) checks it numerically.

Two readings of (b). First, the accuracy of the Laplace approximation is governed by the observations *per cluster*, not by the sample size: a study with 10 000 patients seen twice each is the hard case, and no number of extra patients repairs it. Second, the relevant  $n_i$  is an *effective* one — a cluster of twenty Bernoulli responses says much less about its own  $u_i$  than a cluster of twenty gamma responses — which is why the same approximation behaves so differently in [Example \(Three methods on clustered binary data\)](#) and [Example \(Grunfeld's firms, again\)](#).

Part (c) matters because plain Gauss–Hermite quadrature places its nodes where the *prior* is large. When the data are informative the posterior of  $u_i$  sits elsewhere and is much narrower, so most nodes fall where the integrand is negligible. The adaptive version puts them at each cluster's own mode and curvature — the same  $\hat{u}_i$  and  $\hat{\sigma}_i$  that [Proposition 40.2.3](#) predicts the

random effect with — at the cost of one Newton iteration per cluster per likelihood evaluation.

```

import numpy as np
from numpy.polynomial.hermite import hermgauss
from scipy.optimize import minimize, minimize_scalar

BETA = np.array([-0.5, 1.0])
TAU = 1.0

def expit(z):
    return 1.0 / (1.0 + np.exp(-z))

def cluster_mode(y, X, beta, tau):
    """Mode and curvature of log f(y_i | u) + log p(u), cluster by cluster."""
    eta0, u = X @ beta, np.zeros(y.shape[0])
    for _ in range(100):
        mu = expit(eta0 + u[:, None])
        step = ((y - mu).sum(1) - u / tau**2) / (-(mu * (1 - mu)).sum(1) - 1 / tau
**2)
        u = u - step
        if np.max(np.abs(step)) < 1e-12:
            break
    mu = expit(eta0 + u[:, None])
    return u, 1.0 / np.sqrt((mu * (1 - mu)).sum(1) + 1 / tau**2)

def loglik(y, X, beta, tau, K, adapt=True):
    """Integrated log-likelihood by K-node Gauss-Hermite quadrature.

    Adaptive: centre and scale the nodes at each cluster's own mode and curvature.
    K = 1 with adapt=True is exactly the Laplace approximation."""
    node, w = hermgauss(K)
    if adapt:
        centre, scale = cluster_mode(y, X, beta, tau)
    else:
        centre, scale = np.zeros(y.shape[0]), np.full(y.shape[0], tau)
    pts = centre[:, None] + np.sqrt(2) * scale[:, None] * node # (m, K)
    eta = (X @ beta)[:, :, None] + pts[:, None, :] # (m, n, K)
    )
    cond = (y[:, :, None] * eta - np.logaddexp(0.0, eta)).sum(1) # (m, K)
    prior = -0.5 * (pts / tau) ** 2 - np.log(tau) - 0.5 * np.log(2 * np.pi)
    terms = np.log(w) + node**2 + cond + prior
    return float(np.sum(np.log(np.sqrt(2) * scale) + np.logaddexp.reduce(terms, axis
=1)))

def fit_ml(y, X, K, start=None):
    p = X.shape[2]
    obj = lambda par: -loglik(y, X, par[:p], np.exp(par[p]), K)
    x0 = np.r_[np.zeros(p), 0.0] if start is None else start
    r = minimize(obj, x0, method="Nelder-Mead",
        options=dict(maxiter=8000, maxfev=8000, xatol=1e-8, fatol=1e-10))
    return r.x[:p], float(np.exp(r.x[p])), -r.fun

M_demo, N_demo = 60, 5
rng_demo = np.random.default_rng(40_007)
x_demo = rng_demo.normal(size=(M_demo, N_demo))
X_demo = np.stack([np.ones((M_demo, N_demo)), x_demo], axis=2)
u_demo = rng_demo.normal(0, TAU, M_demo)
y_demo = (rng_demo.uniform(size=(M_demo, N_demo))

```

```

        < expit(X_demo @ BETA + u_demo[:, None])).astype(float)

for K in [1, 5, 15]:
    b, t_hat, ll = fit_ml(y_demo, X_demo, K)
    print(f"AGHQ K = {K:2d}:  beta = {b.round(4)}    tau = {t_hat:.4f}    loglik = {ll:.4f}")

```

[Run this code online](#)

## 40.3.2 Penalized quasi-likelihood

The oldest practical method avoids the integral by treating the random effects as parameters with a penalty.

### Proposition 40.3.2. Penalized quasi-likelihood

In Definition 40.2.1 let  $\ell(\boldsymbol{\beta}, \mathbf{u})$  be the conditional log-likelihood of the data given  $\mathbf{u}$ , and consider the **penalized quasi-likelihood**

$$\ell^{\text{pen}}(\boldsymbol{\beta}, \mathbf{u}) = \ell(\boldsymbol{\beta}, \mathbf{u}) - \frac{1}{2} \mathbf{u}^\top \mathbf{G}^{-1} \mathbf{u}. \quad (40.3.4)$$

- (a) For fixed  $\boldsymbol{\theta}$ , the stationary equations of Eq. 40.3.4 are

$$\mathbf{X}^\top \mathbf{W} \mathbf{D}^{-1} (\mathbf{y} - \boldsymbol{\mu}) = \mathbf{0}, \quad \mathbf{Z}^\top \mathbf{W} \mathbf{D}^{-1} (\mathbf{y} - \boldsymbol{\mu}) = \phi \mathbf{G}^{-1} \mathbf{u},$$

with  $\mathbf{W}$  and  $\mathbf{D}$  the working weights and derivatives of Theorem 34.3.1 evaluated at  $\boldsymbol{\eta} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{u}$ . Equivalently, with the working response  $\mathbf{z} = \boldsymbol{\eta} + \mathbf{D}^{-1}(\mathbf{y} - \boldsymbol{\mu})$  of Eq. 34.4.1, they are Henderson's mixed-model equations (Theorem 32.3.2) for the linear mixed model  $\mathbf{z} \mid \mathbf{u} \sim N_n(\mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{u}, \phi \mathbf{W}^{-1})$ ,  $\mathbf{u} \sim N_q(\mathbf{0}, \mathbf{G})$ .

- (b) Iterating — update  $(\boldsymbol{\beta}, \mathbf{u})$  by those equations, then update  $\boldsymbol{\theta}$  by maximum likelihood or REML in the working linear mixed model, and repeat — is the Breslow–Clayton PQL algorithm. It is therefore the Laplace approximation Eq. 40.3.2 with the further simplification that  $\mathbf{W}$  is treated as fixed when  $\boldsymbol{\theta}$  is updated, so that  $c_i$  contributes no derivative terms.
- (c) The resulting  $\hat{\boldsymbol{\beta}}$  and  $\hat{\boldsymbol{\theta}}$  are biased towards zero, and the bias does **not** vanish as  $m \rightarrow \infty$  with cluster sizes fixed: PQL is inconsistent. It is worst for binary responses with small  $n_i$  and large  $\tau$ , and negligible for responses that carry much information per observation.

*Proof.*

- (a) Differentiating  $\ell$  with respect to  $\boldsymbol{\beta}$  and  $\mathbf{u}$  reproduces Eq. 34.3.1 with model matrix  $[\mathbf{X}, \mathbf{Z}]$ , because  $\mathbf{u}$  enters the linear predictor exactly as  $\boldsymbol{\beta}$  does; the penalty contributes  $-\mathbf{G}^{-1}\mathbf{u}$  to the second block and nothing to the first. Multiplying by  $\phi$  gives the displayed pair. For the second form,  $\mathbf{W}\mathbf{D}^{-1}(\mathbf{y} - \boldsymbol{\mu}) = \mathbf{W}(\mathbf{z} - \boldsymbol{\eta})$  by the definition of  $\mathbf{z}$ , so the equations read  $\mathbf{X}^\top \mathbf{W}(\mathbf{z} - \mathbf{X}\boldsymbol{\beta} - \mathbf{Z}\mathbf{u}) = \mathbf{0}$  and  $\mathbf{Z}^\top \mathbf{W}(\mathbf{z} - \mathbf{X}\boldsymbol{\beta} - \mathbf{Z}\mathbf{u}) = \phi \mathbf{G}^{-1} \mathbf{u}$ , which are Eq. 32.3.5 with  $\mathbf{R} = \phi \mathbf{W}^{-1}$ .
- (b) is the observation that maximizing Eq. 40.3.4 over  $\mathbf{u}$  locates  $\hat{\mathbf{u}}$  at the mode of the integrand of Eq. 40.2.6, which is the first ingredient of Eq. 40.3.2; the remaining ingredient,  $\log|c_i|$ , is what the  $\boldsymbol{\theta}$  update drops.

- (c) We do not prove this. The asymptotic bias expansions, and corrections built from them, are given by Breslow and Lin (1995) and Lin and Breslow (1996); [Example \(Three methods on clustered binary data\)](#) measures the bias by simulation. ■

*Remark* (PQL is empirical Bayes). Since  $\exp\{\ell^{\text{pen}}\}$  is proportional to the joint density of  $\mathbf{y}$  and  $\mathbf{u}$ , the PQL estimate of  $\mathbf{u}$  is the posterior mode of Proposition 40.2.3 with  $\boldsymbol{\beta}$  and  $\boldsymbol{\theta}$  replaced by estimates: an empirical Bayes predictor, and nothing about it is wrong. What is wrong is using the same objective to estimate  $\boldsymbol{\theta}$ , because it has been maximized over  $\mathbf{u}$  rather than integrated over it, and the two differ by exactly the  $\log c_i$  term that carries the information about  $\tau$ .

### 40.3.3 Monte Carlo alternatives

When  $q_0$  exceeds two or three, a product grid needs  $K^{q_0}$  nodes and stops being practical. **Monte Carlo EM** treats  $\mathbf{u}$  as missing data: the E step's expectation over the posterior of  $\mathbf{u}$  is replaced by an average over Markov chain Monte Carlo draws, and the M step is an ordinary weighted GLM fit. EM's monotonicity survives only up to Monte Carlo error, so the number of draws must grow with the iterations; [Chapter 41](#) develops the algorithm and proves that the exact version never decreases the likelihood. **Simulated maximum likelihood** estimates each factor of [Eq. 40.3.1](#) by importance sampling with the Laplace approximation as proposal — [Eq. 40.3.3](#) with random nodes. **Fully Bayesian inference** samples  $\boldsymbol{\beta}$ ,  $\mathbf{u}$  and  $\boldsymbol{\theta}$  together, as [Section 32.6](#) did for the linear mixed model (Proposition 32.6.1); without conjugacy the full conditionals need Metropolis–Hastings steps or data augmentation, which [Chapter 39](#) builds (Definition 39.4.2).

### 40.3.4 How different are the answers?

#### Example. Three methods on clustered binary data

Simulate 150 clusters of 5 binary observations from a random-intercept logistic model with  $\boldsymbol{\beta} = (-0.5, 1.0)^\top$ , a covariate varying within the cluster, and  $\tau = 1$ . Over 200 replications the average estimates are

|                                    | $\hat{\beta}_1$ (truth 1.0) | $\hat{\tau}$ (truth 1.0) |
|------------------------------------|-----------------------------|--------------------------|
| PQL                                | 0.8970                      | 0.8168                   |
| Laplace                            | 1.0022                      | 0.9353                   |
| adaptive Gauss–Hermite,<br>$K = 9$ | 1.0087                      | 0.9909                   |

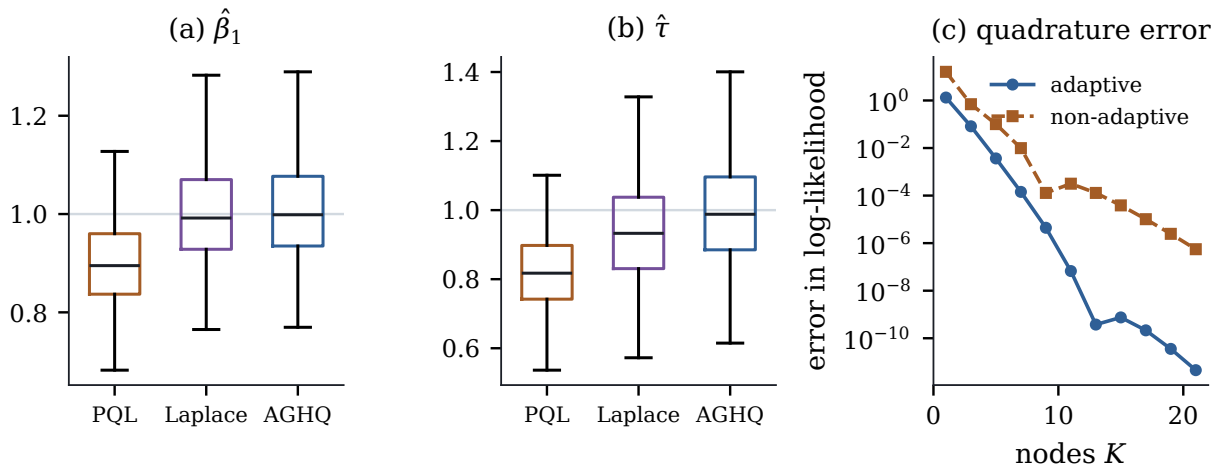


Figure 40.3.1: Approximations to the integrated likelihood, on clustered binary data with  $m = 150$ ,  $n_i = 5$  and  $\tau = 1$ . (a), (b) Sampling distributions of  $\hat{\beta}_1$  and  $\hat{\tau}$  over 200 replications under three methods; the horizontal line is the truth. (c) Error in the integrated log-likelihood against the number of quadrature nodes, adaptive and plain.

PQL attenuates the slope by 10.3 per cent and the standard deviation by 18.3 per cent. The Laplace approximation is nearly unbiased for the slope but still shrinks  $\hat{\tau}$  by about six per cent, which is consistent with the  $O(n_i^{-1})$  error of [Theorem 40.3.1\(b\)](#) at  $n_i = 5$ ; the direction and size of the induced bias are measured here, not derived, since part (b) bounds the error in the likelihood and says nothing about where its maximizer moves. Nine adaptive nodes leave a bias of about one per cent, which is the ordinary finite-sample bias of maximum likelihood rather than a numerical error.

Panel (c) of [Figure 40.3.1](#) separates the numerical question from the statistical one. At a fixed parameter value, the error in the integrated log-likelihood with  $K = 5$  is  $3.65136 \times 10^{-3}$  for the adaptive rule and  $1.01404 \times 10^{-1}$  for the plain one — a factor of about 28 — and the adaptive rule is down to  $3.759 \times 10^{-10}$  by  $K = 13$ , while the plain one is still above  $10^{-6}$ , at  $2.460 \times 10^{-6}$ , when  $K = 19$ .

```
def weighted_lmm(z, X, W):
    """REML fit of a weighted random-intercept linear mixed model:  $z_{ij} = x_{ij}'b + u_i + e_{ij}$ 
    with  $\text{Var}(e_{ij}) = 1/W_{ij}$  and  $\text{Var}(u_i) = t2$ . Returns  $b$ ,  $t2$  and the predicted  $u$ ."""
    def solve(t2):
        s, WX = W.sum(1), W[...] * X
        den = 1 + t2 * s
        XtVX = (np.einsum("mnp,mnq->pq", WX, X)
                 - t2 * np.einsum("mp,mq->pq", WX.sum(1), WX.sum(1) / den[:, None]))
        XtVz = (np.einsum("mnp,mn->p", WX, z)
                 - t2 * np.einsum("mp,m->p", WX.sum(1), (W * z).sum(1) / den))
        return np.linalg.solve(XtVX, XtVz), XtVX, den

    def neg_reml(log_t2):
        t2 = np.exp(log_t2)
        b, XtVX, den = solve(t2)
        r = z - X @ b
        q = np.sum(W * r * r) - t2 * np.sum((W * r).sum(1) ** 2 / den)
        ld = np.sum(np.log(den)) - np.sum(np.log(W))
        return 0.5 * (ld + np.linalg.slogdet(XtVX)[1] + q)
```

```

opt = minimize_scalar(neg_reml, bounds=(-12.0, 6.0), method="bounded",
                      options=dict(xatol=1e-10))
t2 = float(np.exp(opt.x))
b, _, den = solve(t2)
r = z - X @ b
return b, t2, t2 * (W * r).sum(1) / den

def fit_pql(y, X, maxit=300):
    """Breslow-Clayton penalized quasi-likelihood: a linear mixed model on the
    working
    response, refitted until it stops changing."""
    beta, u = np.zeros(X.shape[2]), np.zeros(y.shape[0])
    for _ in range(maxit):
        eta = X @ beta + u[:, None]
        mu = expit(eta)
        v = np.clip(mu * (1 - mu), 1e-8, None)          # working weights
        z = eta + (y - mu) / v                          # working response
        b_new, t2, u_new = weighted_lmm(z, X, v)
        done = max(np.max(np.abs(b_new - beta)), np.max(np.abs(u_new - u))) < 1e-10
        beta, u = b_new, u_new
        if done:
            break
    return beta, float(np.sqrt(t2)), u

b_pql, t_pql, _ = fit_pql(y_demo, X_demo)
print(f"PQL          : beta = {b_pql.round(4)}    tau = {t_pql:.4f}")

```

[Run this code online](#)

The contrast with Example (Grunfeld’s firms, again) is instructive. That fit has  $n_i = 20$  continuous responses per firm, and refitting it with one adaptive node instead of nine changes no parameter by more than  $5 \times 10^{-4}$  and the maximized log-likelihood, which is  $-937.100$ , by less than 0.01. What separates the two is not the software but the information each cluster carries about its own random effect, exactly as [Theorem 40.3.1\(b\)](#) says.

*Idea.* Report which approximation was used. “A random-intercept logistic model was fitted” is not a complete description when PQL and adaptive quadrature can differ by a tenth of the coefficient. With binary responses in small clusters, use quadrature with enough nodes that the answer stops moving, and say how many.

## 40.3.5 Exercises

### 40.3.5.1 A. Check your understanding

#### Exercise. A1

A GLMM with two correlated random effects per cluster is fitted by non-adaptive Gauss–Hermite quadrature on a product grid. How many likelihood evaluations per cluster does  $K = 20$  cost? What if there are four random effects per cluster? What does this say about crossed random effects, where  $q_0$  is effectively the whole vector?

**Exercise. A2**

For each of the following, say whether PQL is likely to be adequate and why: (a) counts with means in the hundreds, ten per cluster; (b) binary responses, three per cluster, with a large estimated  $\tau$ ; (c) grouped binomial responses with  $N_i = 200$  trials, one per cluster.

**40.3.5.2 B. Practice****Exercise. B1**

Verify [Theorem 40.3.1\(c\)](#) for  $K = 1$  directly: show that the one-point Gauss–Hermite rule has node  $z_1 = 0$  and weight  $w_1 = \sqrt{\pi}$ , and that [Eq. 40.3.3](#) then reduces to [Eq. 40.3.2](#) without its correction terms. Then explain why the *non*-adaptive rule with  $K = 1$  gives something quite different, and say what.

*Solution.* The one-point rule must integrate  $e^{-z^2}P(z)$  exactly for  $P$  of degree  $\leq 1$ . Taking  $P = 1$  gives  $w_1 = \int e^{-z^2} dz = \sqrt{\pi}$ ; taking  $P = z$  gives  $w_1 z_1 = \int z e^{-z^2} dz = 0$ , so  $z_1 = 0$ . Substituting in [Eq. 40.3.3](#) gives  $\sqrt{2}\hat{\sigma}_i \sqrt{\pi} e^{\psi_i(\hat{u}_i)}$ , which is  $\sqrt{2\pi/c_i} e^{\psi_i(\hat{u}_i)}$ . The non-adaptive rule puts its single node at  $u = 0$  with scale  $\tau$ , giving  $\sqrt{2\pi\tau} e^{\psi_i(0)}$ : the integrand evaluated at the *prior* mode with the prior's width, which is the right answer only when the data say nothing.

**Exercise. B2**

For a cluster of  $n$  Bernoulli responses with common success probability  $\pi$  and a random intercept, show that  $c_i = n\pi(1 - \pi) + \tau^{-2}$ . Compare with the gamma case of Example (Grunfeld's firms, again), where  $c_i = n/\phi + \tau^{-2}$ , and use [Theorem 40.3.1\(b\)](#) to explain the difference between the two fits in terms of an effective cluster size.

*Solution.* For the logit,  $\psi_i''(u) = -\sum_j \mu_{ij}(1 - \mu_{ij}) - \tau^{-2}$ , which at a common  $\pi$  is  $-n\pi(1 - \pi) - \tau^{-2}$ . For the gamma with a log link,  $\psi_i''(u) = -\nu \sum_j y_{ij} e^{-\eta_{ij}} - \tau^{-2}$ , which at the mode is about  $-n/\phi - \tau^{-2}$  since  $\nu = 1/\phi$  and  $y_{ij} e^{-\eta_{ij}} \approx 1$ . The Laplace error is governed by  $c_i \tau^2$ , so the effective cluster size is  $n\pi(1 - \pi)\tau^2$  against  $n\tau^2/\phi$ . With  $n = 5$ ,  $\pi = 1/2$ ,  $\tau = 1$  the first is 1.25; with  $n = 20$ ,  $\phi = 0.0930$ ,  $\tau = 0.52$  the second is about 58. The two fits are in completely different regimes.

**40.3.5.3 C. Going deeper****Exercise. C1**

[Proposition 40.3.2\(c\)](#) records that PQL is inconsistent but does not prove it. Here is the extreme case, in which everything can be done by hand: the limit  $\mathbf{G}^{-1} \rightarrow \mathbf{0}$  of [Eq. 40.3.4](#), where the penalty disappears and the cluster levels become free parameters. Take matched pairs:  $m$  clusters of size two,  $x_{i1} = 0$ ,  $x_{i2} = 1$ , and  $\text{logit Pr}(Y_{ij} = 1 \mid u_i) = u_i + \beta x_{ij}$ , with  $u_1, \dots, u_m$  and  $\beta$  all maximized over.

(a) Show that a pair with  $y_{i1} + y_{i2} = 0$  or 2 has profile log-likelihood  $\sup_u \ell_i(u, \beta) = 0$ , free

of  $\beta$ , the supremum being approached as  $u \rightarrow \mp\infty$ . Such pairs carry no information about  $\beta$  here.

- (b) Show that a discordant pair has  $\hat{u}_i = -\beta/2$ . (Use  $h(a) + h(b) = 1$  if and only if  $a + b = 0$ , where  $h$  is the logistic inverse link.) Show that its profile log-likelihood is  $\pm\beta/2 - 2\log\{2\cosh(\beta/4)\}$ , the sign being  $+$  for  $(y_{i1}, y_{i2}) = (0, 1)$ .
- (c) Let  $a$  and  $b$  count the discordant pairs of the two kinds. Show that the joint maximizer satisfies  $\tanh(\hat{\beta}/4) = (a - b)/(a + b)$ , equivalently  $h(\hat{\beta}/2) = a/(a + b)$ .
- (d) Show that  $\Pr(Y_{i2} = 1 \mid Y_{i1} + Y_{i2} = 1) = h(\beta)$ , free of  $u_i$ , so that the conditional maximum likelihood estimate  $\tilde{\beta}$  solves  $h(\tilde{\beta}) = a/(a + b)$  and is consistent. Conclude that  $\hat{\beta} = 2\tilde{\beta} \rightarrow 2\beta$  as  $m \rightarrow \infty$ .

The bias does not vanish with the number of clusters. In which direction is it, and how does that sit with [Proposition 40.3.2\(c\)](#)?

*Solution.*

- (a) For  $(0, 0)$ ,  $\ell_i(u, \beta) = -\log(1 + e^u) - \log(1 + e^{u+\beta})$ , which is negative and tends to 0 as  $u \rightarrow -\infty$ ; for  $(1, 1)$ ,  $\ell_i = 2u + \beta - \log(1 + e^u) - \log(1 + e^{u+\beta}) \rightarrow 0$  as  $u \rightarrow +\infty$ . In both cases the supremum is 0 whatever  $\beta$  is.
- (b) For  $(0, 1)$ ,  $\ell_i = (u + \beta) - \log(1 + e^u) - \log(1 + e^{u+\beta})$  and  $\partial\ell_i/\partial u = 1 - h(u) - h(u + \beta)$ , which vanishes when  $h(u) + h(u + \beta) = 1$ , that is when  $u + (u + \beta) = 0$ , since  $h(b) = 1 - h(a)$  means  $h(b) = h(-a)$  and  $h$  is injective. So  $\hat{u}_i = -\beta/2$ , and the same computation applies to  $(1, 0)$ . Substituting,  $(1 + e^{-\beta/2})(1 + e^{\beta/2}) = 2 + 2\cosh(\beta/2) = 4\cosh^2(\beta/4)$ , so the profile value is  $\beta/2 - 2\log\{2\cosh(\beta/4)\}$  for  $(0, 1)$  and  $-\beta/2 - 2\log\{2\cosh(\beta/4)\}$  for  $(1, 0)$ .
- (c) Adding (a) and (b), the profile log-likelihood is  $(a - b)\beta/2 - 2(a + b)\log\{2\cosh(\beta/4)\}$ . Differentiating,  $(a - b)/2 - \frac{1}{2}(a + b)\tanh(\beta/4) = 0$ , which gives  $\tanh(\hat{\beta}/4) = (a - b)/(a + b)$ . Since  $h(\beta/2) = \{1 + \tanh(\beta/4)\}/2$ , this is  $h(\hat{\beta}/2) = a/(a + b)$ . (The function is strictly concave, so the stationary point is the maximum.)
- (d) The two discordant outcomes have conditional probabilities proportional to  $\{1 - h(u)\}h(u + \beta)$  and  $h(u)\{1 - h(u + \beta)\}$ , whose ratio is  $e^{u+\beta}/e^u = e^\beta$  — the  $u$  cancels. So  $\Pr(Y_{i2} = 1 \mid \text{total} = 1) = h(\beta)$ , the conditional likelihood is  $h(\beta)^a\{1 - h(\beta)\}^b$ , and its maximizer solves  $h(\tilde{\beta}) = a/(a + b)$ . The pairs are independent, so  $a/(a + b) \rightarrow h(\beta)$  with probability one and  $\tilde{\beta} \rightarrow \beta$ . Comparing with (c),  $h(\hat{\beta}/2) = h(\tilde{\beta})$ , so  $\hat{\beta} = 2\tilde{\beta} \rightarrow 2\beta$ .

The bias here is *away* from zero, by a factor of two, and it is the incidental-parameter effect: one nuisance parameter per cluster, estimated from two observations, and never consistently. PQL as it is actually used keeps the penalty  $-\frac{1}{2}\mathbf{u}^\top \mathbf{G}^{-1}\mathbf{u}$ , which pulls the  $\hat{u}_i$  towards zero and with them  $\hat{\beta}$ ; [Proposition 40.3.2\(c\)](#) and the simulation of [Example \(Three methods on clustered binary data\)](#) report that in a random-intercept logistic model with small clusters the shrinkage wins, so the net bias is towards zero. The two effects are different and they need not cancel, which is why the size of the PQL bias is not predictable from  $m$  and  $n_i$  alone.

### Exercise. C2

The correction  $\delta_i$  of [Eq. 40.3.2](#) can be computed in closed form in two instructive cases.

- (a) Take  $\psi(u) = mu - e^u$  with  $m > 0$ . Show that  $\int_{\mathbb{R}} e^{\psi(u)} du = \Gamma(m)$ , that  $\hat{u} = \log m$ ,

$c = m$  and  $\psi'''(\hat{u}) = \psi^{(4)}(\hat{u}) = -m$ , and hence that  $\delta = 1/(12m)$ .

- (b) Deduce that [Eq. 40.3.2](#) reads  $\Gamma(m) = \sqrt{2\pi} m^{m-1/2} e^{-m} \{1 + 1/(12m) + \dots\}$ , which is Stirling's series: the expansion of [Theorem 40.3.1\(b\)](#) is a familiar object in disguise.
- (c) Now take a cluster of  $n$  Poisson responses with a log link, a common linear predictor  $\eta$  and a random intercept of variance  $\tau^2$ , so that  $\psi_i(u) = S(\eta + u) - ne^{\eta+u} - u^2/(2\tau^2) + \text{const}$  with  $S = \sum_j y_{ij}$ . Writing  $M = ne^{\eta+\hat{u}_i}$ , show that  $c_i = M + \tau^{-2}$ ,  $\psi_i'''(\hat{u}_i) = \psi_i^{(4)}(\hat{u}_i) = -M$ , and

$$\delta_i = \frac{M(5M - 3c_i)}{24c_i^3} = \frac{M(2M - 3\tau^{-2})}{24(M + \tau^{-2})^3}.$$

- (d) Deduce that  $\delta_i > 0$  exactly when  $M\tau^2 > \frac{3}{2}$ , and that  $\delta_i \rightarrow 1/(12M)$  as  $M \rightarrow \infty$  with  $\tau$  fixed. Relate  $M\tau^2$  to the effective cluster size of [Exercise \(B2\)](#).

*Solution.*

- (a) The substitution  $t = e^u$  gives  $\int e^{mu-e^u} du = \int_0^\infty t^{m-1} e^{-t} dt = \Gamma(m)$ . Then  $\psi'(u) = m - e^u$  vanishes at  $\hat{u} = \log m$ ;  $\psi''(u) = -e^u$ , so  $c = -\psi''(\hat{u}) = m$ ; and every higher derivative is  $-e^u = -m$  at the mode. Substituting in [Eq. 40.3.2](#),

$$\delta = \frac{-m}{8m^2} + \frac{5m^2}{24m^3} = \frac{-3 + 5}{24m} = \frac{1}{12m}.$$

- (b)  $\psi(\hat{u}) = m \log m - m$ , so  $\sqrt{2\pi/c} e^{\psi(\hat{u})} = \sqrt{2\pi/m} m^m e^{-m} = \sqrt{2\pi} m^{m-1/2} e^{-m}$ , and [Eq. 40.3.2](#) multiplies it by  $1 + \delta + \dots = 1 + 1/(12m) + \dots$ . That is the classical expansion of the gamma function, whose correctness is known independently — a check on the formula, and a reminder that the Laplace approximation of this section is the Stirling approximation with the integrand changed.
- (c)  $\psi_i'(u) = S - ne^{\eta+u} - u/\tau^2$  and  $\psi_i''(u) = -ne^{\eta+u} - \tau^{-2}$ , so  $c_i = -\psi_i''(\hat{u}_i) = M + \tau^{-2}$ ; the Gaussian term contributes nothing beyond the second derivative, so  $\psi_i''' = \psi_i^{(4)} = -ne^{\eta+u}$ , equal to  $-M$  at the mode. Then

$$\delta_i = \frac{-M}{8c_i^2} + \frac{5M^2}{24c_i^3} = \frac{M(5M - 3c_i)}{24c_i^3},$$

and  $5M - 3c_i = 5M - 3M - 3\tau^{-2} = 2M - 3\tau^{-2}$ .

- (d) The denominator is positive, so the sign of  $\delta_i$  is that of  $2M - 3\tau^{-2}$ , and  $\delta_i > 0$  iff  $M\tau^2 > \frac{3}{2}$ . As  $M \rightarrow \infty$ ,  $c_i \sim M$  and  $\delta_i \sim 2M^2/(24M^3) = 1/(12M)$ , which is the  $O(n_i^{-1})$  of [Theorem 40.3.1\(b\)](#) with its constant exhibited, since  $M$  is proportional to  $n$ . The product  $M\tau^2 = c_i\tau^2 - 1$  is the effective cluster size of [Exercise \(B2\)](#), so a cluster carrying little information about its own random effect —  $M\tau^2$  below  $\frac{3}{2}$  — has a correction of the *opposite* sign, and the Laplace approximation then overstates the integral. The transition sits in exactly the range where [Example \(Three methods on clustered binary data\)](#) found Laplace shrinking  $\hat{\tau}$  and [Example \(Grunfeld's firms, again\)](#) found it faultless.